Project Report: Sovereign Sentry

100% Offline, Native-Hindi Smart Assistant for Edge Devices

1. Abstract

Sovereign Sentry is a fully offline, air-gapped voice automation system designed to run on resource-constrained edge hardware (Raspberry Pi 4 - 4GB RAM). Unlike traditional smart home assistants that rely on cloud APIs and compromise user privacy, this system processes native Hindi speech, executes complex temporal commands (alarms/reminders), and controls local hardware entirely on-device. It utilizes a highly optimized hybrid NLP engine, combining deterministic regex parsers with a natively trained 2-Billion parameter Indian LLM for semantic fallback.

2. System Architecture & Core Modules

The system is divided into highly modular, decoupled components to ensure maximum RAM efficiency and prevent bottlenecking on the Pi's CPU.

A. The Master Controller (`main.py`)

This is the central nervous system that routes audio data, manages background threads, and executes hardware commands.

- **Acoustic Engine (Ears):** Utilizes the `Vosk` KaldiRecognizer loaded with the `vosk-model-small-hi-0.22` acoustic model. It runs dual-recognizers: a lightweight wake-word detector ("सुनो", "नमस्ते") to conserve CPU, and a main recognizer that activates only upon waking.
- **Voice Synthesis (Mouth):** Integrates the `Piper` TTS engine using the highly compressed `hi_IN-pratham-medium.onnx` model. It bypasses heavy Python audio libraries by executing directly via OS subprocesses (`paplay`/`start`), ensuring zero-latency speech generation.
- **Hardware Execution:** Maps parsed intents directly to the local `hardware.py` interface, handling tasks like `LIGHT_ON`, `FAN_OFF`, and real-time OS volume control (`pactl`).

B. The Hybrid Intelligence Engine (`intentparser.py`)

To prevent the 4GB Raspberry Pi from overflowing into swap memory, the cognitive layer is split into a Two-Tier Architecture:

- **Tier 1: Deterministic Fuzzy Matching (RapidFuzz):** Uses the `rapidfuzz` library to compare incoming Hindi text against a titanium-grade `COMMAND_REGISTRY`. If the spoken command matches a known pattern with >60% confidence, the command executes in milliseconds without touching the LLM.
- **Tier 2: Semantic AI Fallback (Sarvam-1 LLM):** If the user speaks heavy, unpredictable slang, the input is passed to the native **Sarvam-1 (2B) GGUF** model. Loaded via `llama_cpp`, the model is strictly confined to a 512-token context window (`n_ctx=512`) to protect the hardware. It reads the context natively in Devanagari and outputs the exact programmatic intent.

C. The Temporal Memory System (`main.py` & `memory.json`)

Sovereign Sentry features a persistent, offline event manager capable of understanding complex human time expressions.

- **Regex Time Extraction:** A custom `re` engine hunts for digit-based formats (e.g., "7:30") and native Hindi fractional times (e.g., "साढ़े", "पौने"). It dynamically calculates exact future `datetime` objects.
- **Name Extraction:** Dynamically isolates the specific name of the event (e.g., "मीटिंग", "दवाई") by stripping out time and date vocabulary from the user's phrase.
- **Persistent Storage (`memory.json`):** Instead of volatile RAM timers, the calculated `datetime` and custom event name are physically written to a local JSON database.
- **The Timekeeper Daemon:** A parallel `threading.Thread` runs silently in the background. It wakes up every 10 seconds (`time.sleep(10)`), compares the current OS time to the JSON database, triggers the `Piper` alarm if a match is found, and automatically deletes the fulfilled event.

3. Key Engineering Achievements

1. **Zero Cloud Dependency:** The entire system—from speech-to-text, intent classification, memory storage, and text-to-speech—requires absolutely zero internet connection, guaranteeing total data sovereignty.
2. **Native Devanagari Tokenization:** By utilizing Sarvam-1 instead of an English-first foundation model (like Llama or Qwen), the system avoids "token bloat" when reading Hindi, significantly reducing the computational load on the Pi's CPU.
3. **RAM Optimization:** The footprint is tightly controlled. With Pi OS (~800MB), Vosk (~300MB), and the quantized Sarvam-1 model (~1.5GB), the system runs comfortably within the 4GB hardware limit without thermal throttling or crashing.
4. **Resilient Memory:** Because alarms and reminders are written to physical storage (`memory.json`), the user's schedule survives unexpected power outages or system reboots.

4. Conclusion

The Sovereign Sentry project successfully demonstrates that high-fidelity, native-language smart home automation is possible on constrained edge devices. By combining lightweight regex rules with a heavily quantized local LLM, the system achieves near-instant response times for daily tasks while retaining the deep semantic intelligence required to parse complex human slang—all while remaining completely invisible to the outside internet.